

Rossmann Store Sales Forecasting

Final Project Report — NextHikes IT Solutions

Objective: build an end-to-end sales forecasting solution for multiple retail stores, including exploratory analysis, machine learning, deep learning and a web interface.

1. Data and Cleaning

The supplied work contains 1,017,209 training rows, 41,088 test rows and 1,115 stores. The cleaning workflow handled missing Open values in test data, missing competition distance, structural Promo2 missing values, and 54 data-error rows where Sales=0 while Open=1. High sales spikes were retained because they can represent genuine promotion or holiday behaviour.

2. Exploratory Findings

Promotions increased average sales from 5,930 to 8,229 and sales per customer from 8.51 to 9.74. Sales and customers had a strong positive correlation ($r=0.824$). Competition distance had a weak raw correlation with sales ($r=-0.042$). Extended/extra assortment stores showed higher average sales, and stores open on Sundays had higher average sales in the supplied analysis.

3. Machine Learning

A time-based validation split was used, holding out the final six weeks to mimic the forecasting requirement. Date, weekday/weekend, month position, competition age, Promo2 duration and promotion-related features were engineered. A scikit-learn Pipeline combined preprocessing and RandomForestRegressor. RMSPE was selected as the primary metric because it measures relative error and matches the Rossmann forecasting context.

Validation results

Metric	Random Forest
RMSPE	0.2425 (24.25%)
MAE	1057.84
RMSE	1529.85

4. Post-Prediction Analysis

Feature importance identified CompetitionDistance, Store, Promo, CompetitionOpenMonths, DayOfWeek and Promo2OpenWeeks among the strongest signals. The project also implements an empirical prediction interval using the spread of individual random-forest tree predictions.

5. Deep Learning — LSTM

The supplied LSTM notebook isolates Store 1 as a daily time series, performs an Augmented Dickey-Fuller stationarity check, examines ACF/PACF, uses a 14-day sliding window, scales values to $(-1,1)$, and trains a two-layer LSTM. The supplied log reports MAE 471.45, RMSE 595.48 and RMSPE 0.1349 on the test period.

6. Deployment

A Streamlit dashboard is included. It supports manual store/date inputs or CSV upload, predicts sales and customers, displays the results in a chart and provides CSV download. MLflow and DVC setup files are included for experiment tracking and data versioning.

7. Conclusion

The project delivers the requested end-to-end workflow: EDA, preprocessing, feature engineering, machine learning, LSTM forecasting, experiment/data versioning hooks and a prediction interface. The remaining submission actions are operational: push the repository to GitHub, run MLflow/DVC in the learner's environment to capture the required screenshots, and deploy the Streamlit application.